Leveraging Large Language Models to Understand the Opinion of Polarization

An Introduction of Large Language Models Empowered Agents and Social Simulation

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Introduction

Opinion Polarization

- Opinion polarization and fragmentation appear increasingly pronounced over various issues [2].
- The mechanisms of opinion polarization are diverse and heterogeneous.
  - Psychological processes (eg., Confirmation Bias[1])
  - External factors (eg., Algorithm bias[5], Media [3])

Problems

Studies on specific events, groups, or contexts may not always be generalizable to the broader real world due to inherent complexity and randomness.

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Agent-based Models (ABMs)

- ABMs are computational models that simulate the interactions of individual agents within a specific environment.
- Each agent follows set rules and can adapt its behavior and characteristics based on its interactions with other agents and the environment itself.

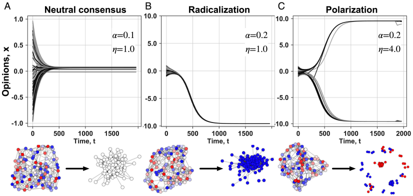

Figure 1: Simulation of opinion dynamics across different scenarios[4]

Agent-based Models (ABMs)

- Conventional ABMs often rely on abstract theories, oversimplified assumptions, and predefined rules.

Limitations in Interpretability and Generalization

- Model Complexity: Simple agent architecture is not enough to cope with complex tasks.
- Agent Heterogeneity and Model Adaptability: Difficult to develop a general agent that can support simulations across environments.
- Model Interpretability: Cannot support integrative simulation in real-world problems.

Large Language Models(LLMs)

- LLMs are typically trained on large-scale human language corpora.
- Fine-tuned through reinforcement learning from human feedback (RLHF)
  - LLMs, with their ability to process and generate natural language, can be used to mimic human behaviors.
  - LLMs possess strong reasoning capabilities, which can be leveraged to improve the model interpretability.

Propmts

- In-context learning(ICL): LLMs can address a new task during inference by receiving a prompt.

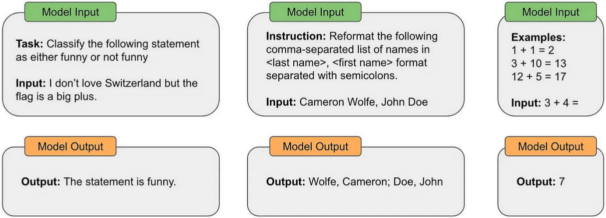

Figure 2: Use prompts to guide LLMs

Motivation

LLMs as Agents

- Create an LLMs-powered agent system, where LLM functions as the agent's brain that complements action, planning, memory, and tool use.

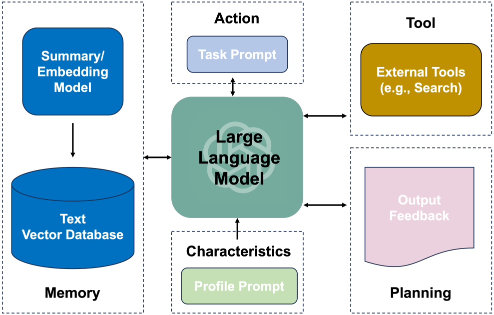

Figure 3: Overview of LLMs-powered agents

LLMs in Multi-Agent Simulation

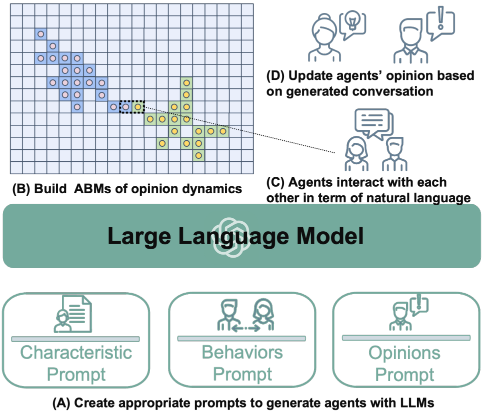

Figure 4: Overview of ABMs powered by LLMs

Case Study

Purpose

Modeling in Natural Language

Instead of relying on empirical rules or mathematical formulas to define agent attributes, LLM-powered agents can process natural language for both input and output.

Alignment with Human Behaviors and Values

LLM-powered agents can be instructed to align with human values and behaviors, including demographic characteristics, cognitive biases, and behavioral tendencies.

Data

Issue	Policy	Biden	Trump
Decrease foreign aid	Decrease U.S. foreign aid spending	Disagree	Agree

Favor Treatment	Against Treatment
Due to corruption and weak institutions ...	The question of whether to decrease foreign ...

Table 1: Policy issues, party leader preference, and persuasive message treatments. 24 contemporary policy issues and 48 persuasive messages containing arguments and evidence [6].

Method

- LLMs-powered agents use the *GPT-3.5-turbo-instruct* model with a sampling temperature of 0.8.
- LLMs-powered agent is required to update opinions toward specific policy after reading the persuasive message.
  - RQ1: Whether persuasive messages in support of the opposing political leader's policy preferences can increase the level of agreement with the previously disagreed-upon policy.
  - RQ2: Whether persuasive messages against the supported political leader's policy preferences can decrease the level of agreement with the previously supported policy.

$$TE_{ij} = \beta_0 + \beta_1 Ideology_i + \epsilon$$

Partisan of LLMs-powered Agents

Prompt of Initialing Agents

Role play this person.
Ideologically, you are {ideology}. Regarding party support, you align with {party}.
Please tell me about your opinions about {issue}.
Additionally, please provide a score to assess your view of {policy}, ranging from 0 (entirely disagree) to 10 (entirely agree).

Partisans of LLMs-powered Agents

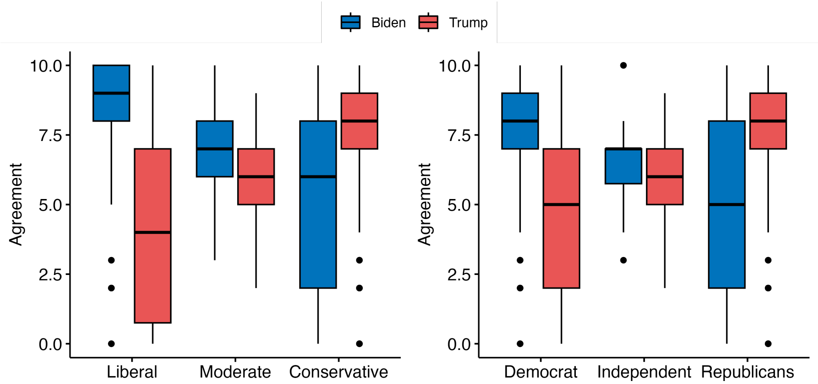

Figure 5: Agreement with policies supported by Biden and Trump across various ideologies and party affiliations

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Update Opinions by LLMs-powered Agent

Prompt of Updating Opinions after Reading Message

You would see a new piece of text about {issue} and be asked to answer your honest opinion about {issue}.
Your current stance on {issue} is summarized as '{history}'.
You are then introduced to the following information::'{text}'
After reading this text, you have the opportunity to reconsider your opinion and stance on {issue}
After the reconsideration, please tell me about your opinions about {issue} and a score to assess your view of {policy}, ranging from 0 (entirely disagree) to 10 (entirely agree).

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Update Opinions with Confirmation Bias by LLMs-powered Agent

Prompt of Updating Opinions with Confirmation Bias after Reading Message

You would see a new piece of text about {issue} and be asked to answer your honest opinion about {issue}...

......

Remember, you are role-playing as a real person. Like humans, you have confirmation bias. You will be more likely to believe information that supports your current stance and less likely to believe information that contradicts your current stance.

......

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How Confirmation Bias Affect Agents' Opinion Changes

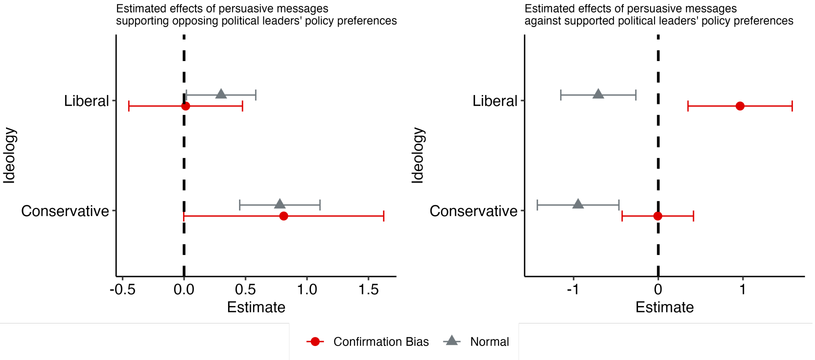

Figure 6: Effects of persuasive messages on *normal* agents ($N = 50$) and agents with confirmation bias ($N = 50$)

Discussion

Discussion

- Minimizes reliance on predefined assumptions and rules and instead leverages built-in knowledge of LLMs to replicate human behaviors and decisions.
- Incorporate real-world data to open up the possibility for social simulation that closely mimics reality.
- The Interpretative nature of LLMs provides more intuitive and human-understandable representations of agents' behaviors and interactions.

Acknowledgements

Acknowledgements

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